

Pon Aung

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Portfolio: PonAppliedPhysics.github.io

EDUCATION

The Ohio State University

Bachelor of Science in Physics

Columbus, OH

May 2026

Minor: Engineering Science, Mechanical Engineering Emphasis

Coursework: Statics, Mechanics of Materials, Dynamics, Material Science, Intermediate Electricity and Magnetism, Experimental Physics Instrumentation, Optics

EXPERIENCE

Honda Development and Manufacturing of America

Marysville, OH

Project Leader – BEV Thermal Runaway Detection Capstone

August 2024 – May 2025

- Narrowed 9 thermal runaway detection technologies to 4 sensor candidates through weighted decision analysis against a stakeholder-driven safety hierarchy for Honda's 800-lb BEV battery packs
- Designed and executed ultrasonic pulse-echo experiments (Olympus V103-SB, 5 V peak-to-peak at 1 MHz) across full charge and discharge cycles, with one-way ANOVA confirming statistically significant voltage variation across battery charge states ($p = 1.91\text{e-}4$ charging, $p = 4.17\text{e-}60$ discharging)
- Tested infrared pyrometer accuracy at 1.5, 3, and 4.5 ft against a FLIR thermal reference, identifying that black surface treatment on aluminum improved detection while reflective surfaces degraded readings beyond 4.5 ft
- Validated HF gas sensor detection at 0.5, 1.5, and 2.5 m with coefficient of variation under 4 percent for timing and under 8 percent for concentration, confirming sub-1 ppm sensitivity at the longest test distance
- Delivered sensor validation data, selection matrix, and layered multi-sensor deployment strategy to Honda engineering leadership, completing all testing at 36 percent of the \$3,500 budget

PROJECT EXPERIENCE

OSU Underwater Robotics – Talos AUV

Columbus, OH

Mechanical Team Member

December 2025 – May 2026

- Designed hull and chassis components in SolidWorks and machined parts on lathe, mill, and 3D printer to integrate a dual stereo-camera vision system into the Talos AUV for the RoboSub competition
- Redesigned torpedo launcher servo housings in Delrin with built-in mechanical seals and a custom gearbox to convert 270-degree servo rotation to full range of motion, eliminating all water-ingress failures from the RoboSub 2024 design
- Integrated mechanical, electrical, and software subsystems through iterative pool testing, adding a structural brace and guiding rod to the torpedo interface that improved launch accuracy over the prior design

Computational Heat Transfer Analysis and Experimental Validation

Columbus, OH

Physics 5700 – Advanced Experimental Physics

Spring 2026

- Validated 11-thermocouple sensor array data by measuring and subtracting per-sensor systematic offsets ($\sim 0.6^\circ\text{C}$), then confirming cross-run consistency between two independent cooling experiments
- Developed 5 progressively refined finite-difference models, reducing chi-squared error by 10x through data-driven boundary conditions and SciPy-optimized radial heat loss; vectorized NumPy implementation cut runtimes from minutes to 3 seconds
- Built a physics-informed neural network in TensorFlow that recovered thermal diffusivity as a trainable parameter directly from sensor data, achieving MAE of 0.42°C without manual parameter tuning

TECHNICAL SKILLS

Software: MATLAB, Python (NumPy, SciPy, Pandas, Matplotlib, TensorFlow), SolidWorks, Shapr3D

Analysis: Finite-difference PDE solvers, parameter optimization, stability analysis, chi-squared goodness-of-fit testing, ANOVA, uncertainty propagation, weighted least-squares regression

Mechanical Engineering: Heat transfer, statics, dynamics, structural analysis, circuits, sensor validation, lab instrumentation and data collection